

AI Opportunity Atlas v1 - Methodology

This document summarizes the AI Opportunity Atlas methodology.

Core Idea:

Each scientific problem is treated as an opportunity node characterized by impact, difficulty, AI solvability, and leverage.

Primary Variables:

Humanity Impact (H): 0-100 measure of societal benefit.

Economic Impact (E): 0-100 measure of economic transformation.

Scientific Difficulty (D): 0-100 measure of hardness.

LLM Progress Likelihood (L): probability AI can make progress in 5-10 years.

Scoring Function:

AI Leverage Score = $(H * L) / D$

Secondary Metrics:

Expected Societal Return = $H * E$

AI Replaceability Index measures how much a problem is search or simulation driven.

Recursive AI Benefit measures whether solving the problem improves AI systems themselves.

Taxonomy:

OP = open problem

TECH = technical paper revealing bottleneck

REV = survey of challenges

ROAD = research roadmap

Final Ranking:

Combination of AI Leverage Score, Expected Societal Return, Recursive AI Benefit, and confidence.

Limitations:

This is a heuristic framework, not a calibrated predictive model. Scores are subjective estimates.